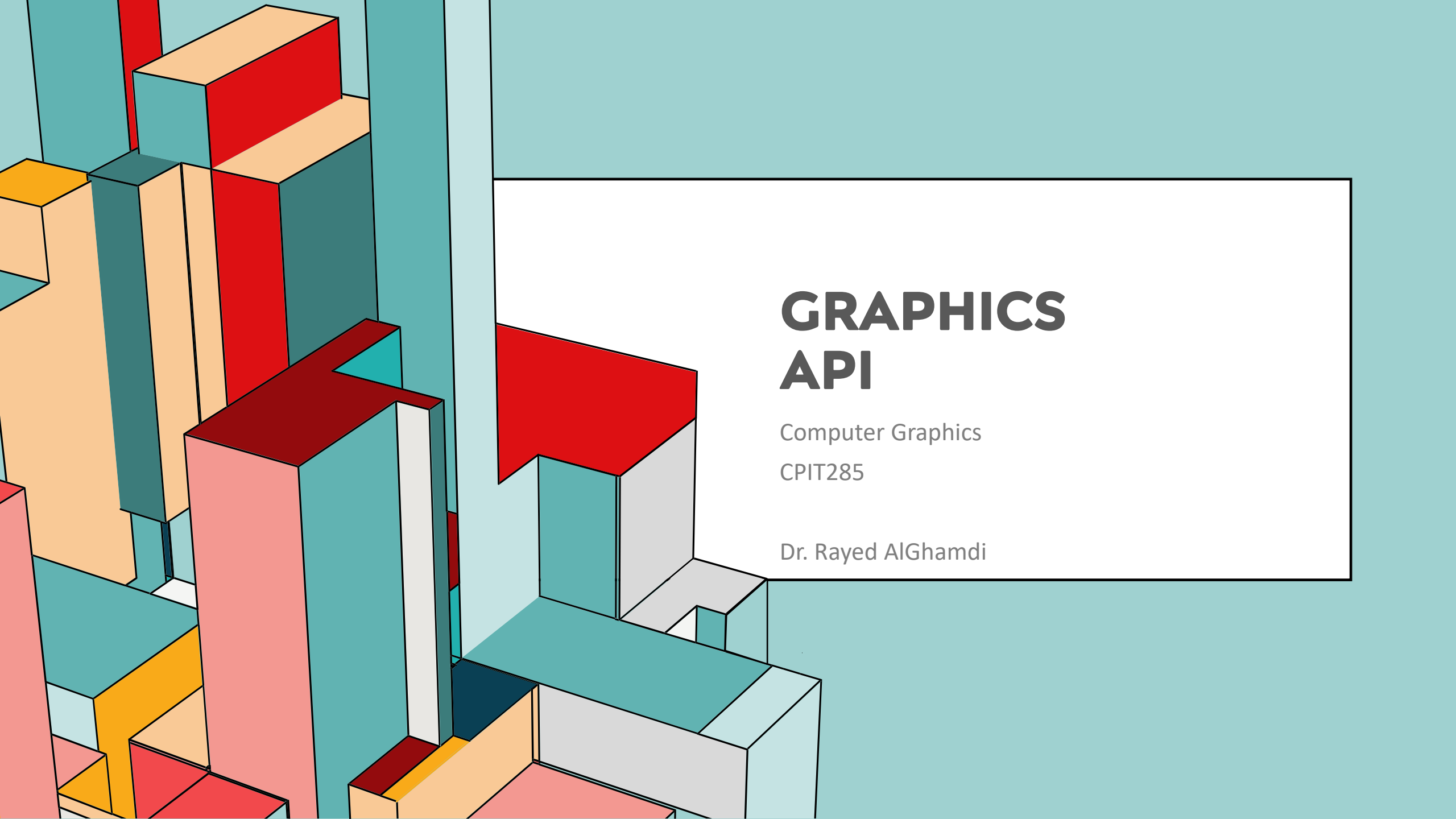




GRAPHICS API

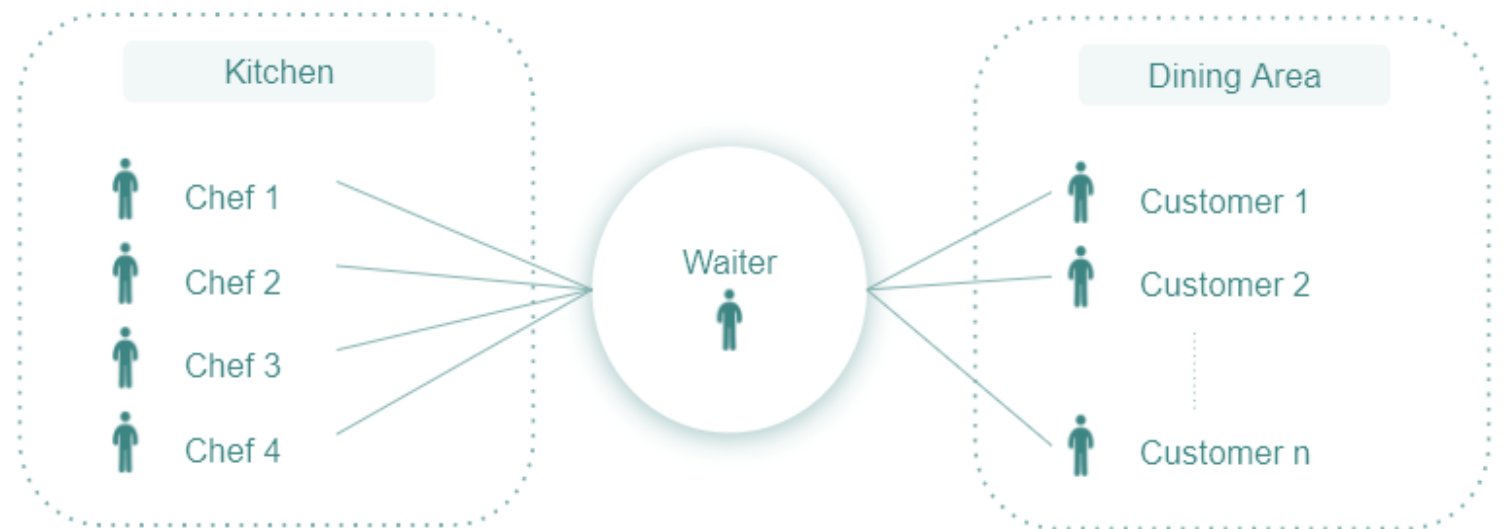
Computer Graphics
CPIT285

Dr. Rayed ALGhamdi

WHAT IS API?

- API is the acronym for Application Programming Interface
- It is an intermediate application that allows different software/services to communicate and serve each other

When you are at restaurant to order meal, it is hard to enter the kitchen to explore what is available there! The kitchen here is a part of the “system” that will prepare your order. What is missing here is the link to communicate and serve your order. That’s where the waiter or API comes in. The waiter is the key intermediate that takes your order & tells the kitchen/system what to do, then serves you back.



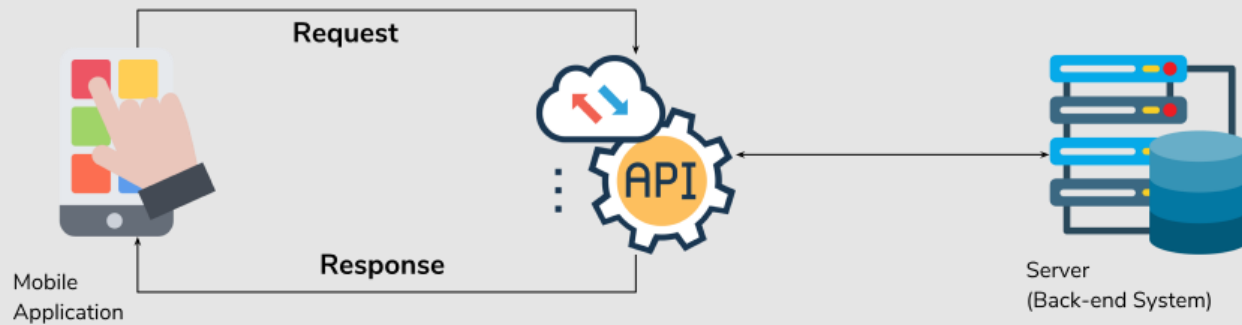
RESTAURANT EXPERIENCE

Back again to restaurants ☺

- You almost have similar steps since you make your order till you are served in any restaurant you visit in the world
- You do not need to have a special manual for every restaurant...



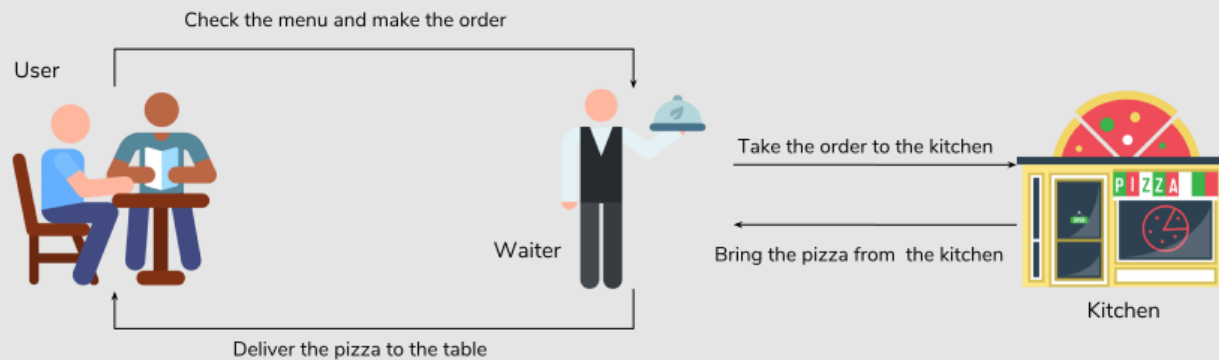
WHAT IS API?



API is a set of definitions that determines how one set of software communicates with another set of software

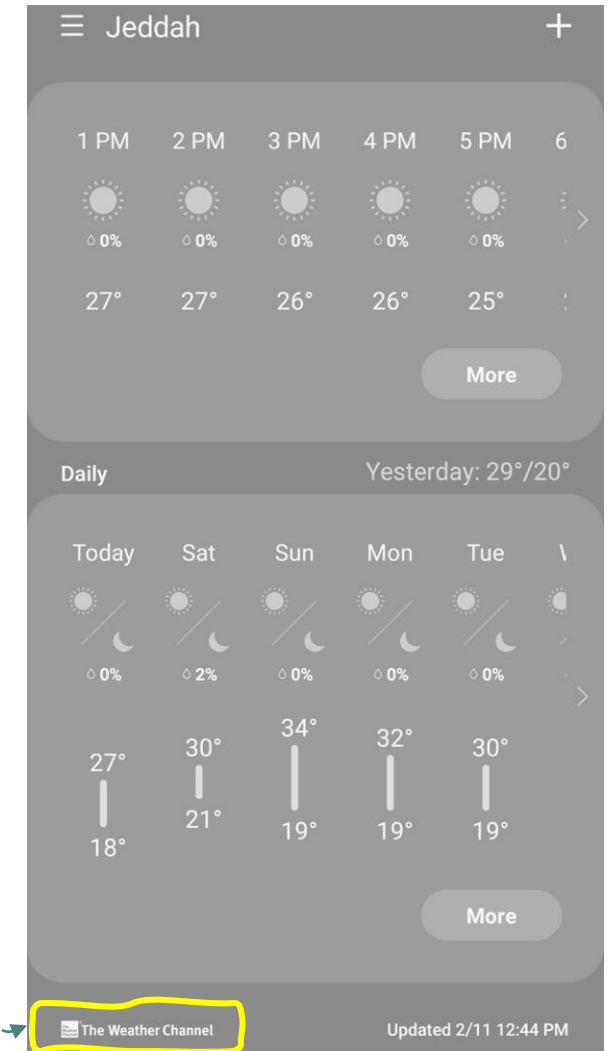
API

- Standardize
- Abstract
- Communicate



WHAT IS API?

- Think of a weather APP on your smartphone
 - Does the App developer/s use tools and types of equipment for measuring and predicting weather conditions?
 - No, it is not! They use APIs provided as a service by climate centers to retrieve weather predictions...
- Think about yourself as a programmer
 - When you want to include maps as service in your App, what to do?
 - Do you design your own map?!
 - Or simply use Google maps?
 - How to allow your App communicate with google maps?
 - You need to travel to USA, meet Google CEO and get his permission to access their maps 😊



+

○

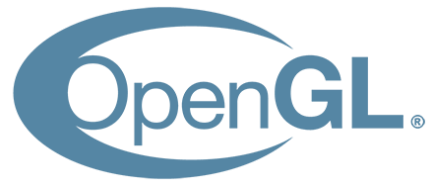
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GRAPHICS API

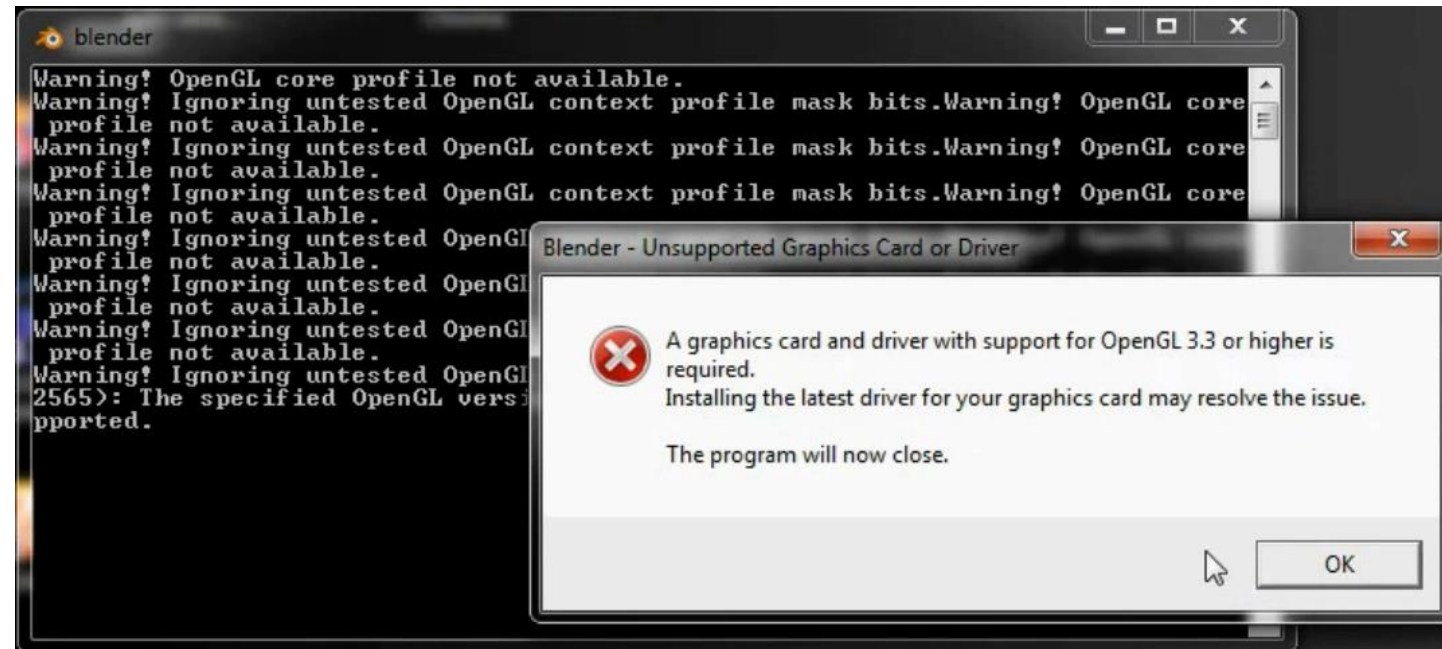


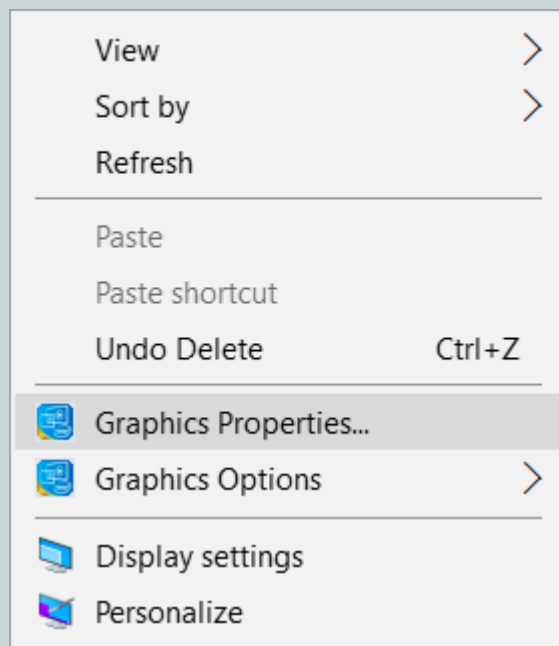
GRAPHICS API

- Think now about graphics APIs.
- Major operating systems we are using today are Graphical User Interface (GUI).
- Without a graphics API, you will end up using **none** GUI-based operating systems like DOS!
- The graphics APIs come by default with the GUI operating systems.
- In case you encounter any problem or require an updated version of the graphics API on your computer, updating the graphics card driver/operating system fixes the problem.

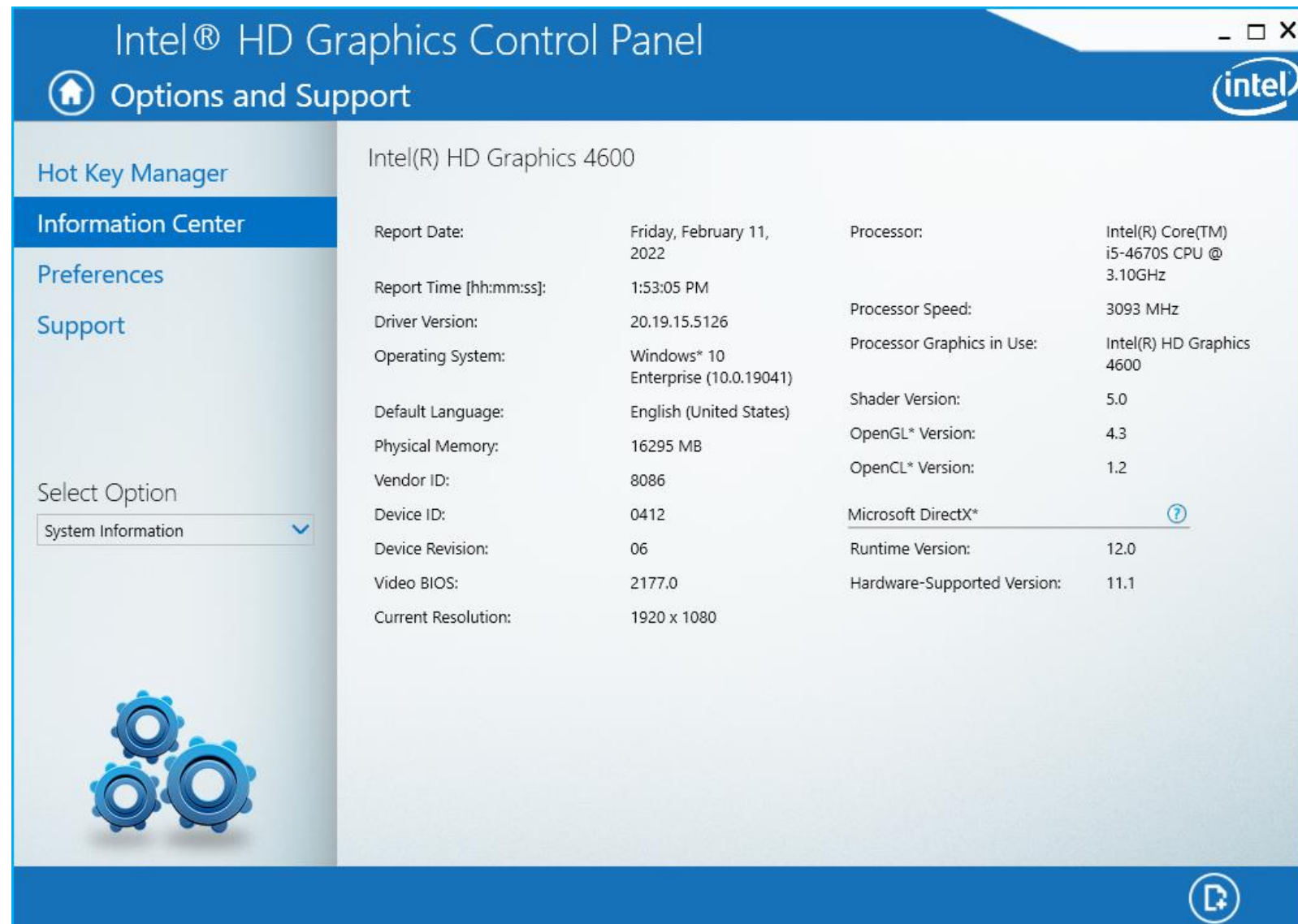


Have you faced a similar problem when you tried to run a graphics application such as a blender or a game? A popup message tells you the current version of OpenGL or Direct X is not supported. OpenGL in this case is the API/ the intermediate between the application and the GPU driver. There is no link for the app to communicate with the GPU when the API is missing!

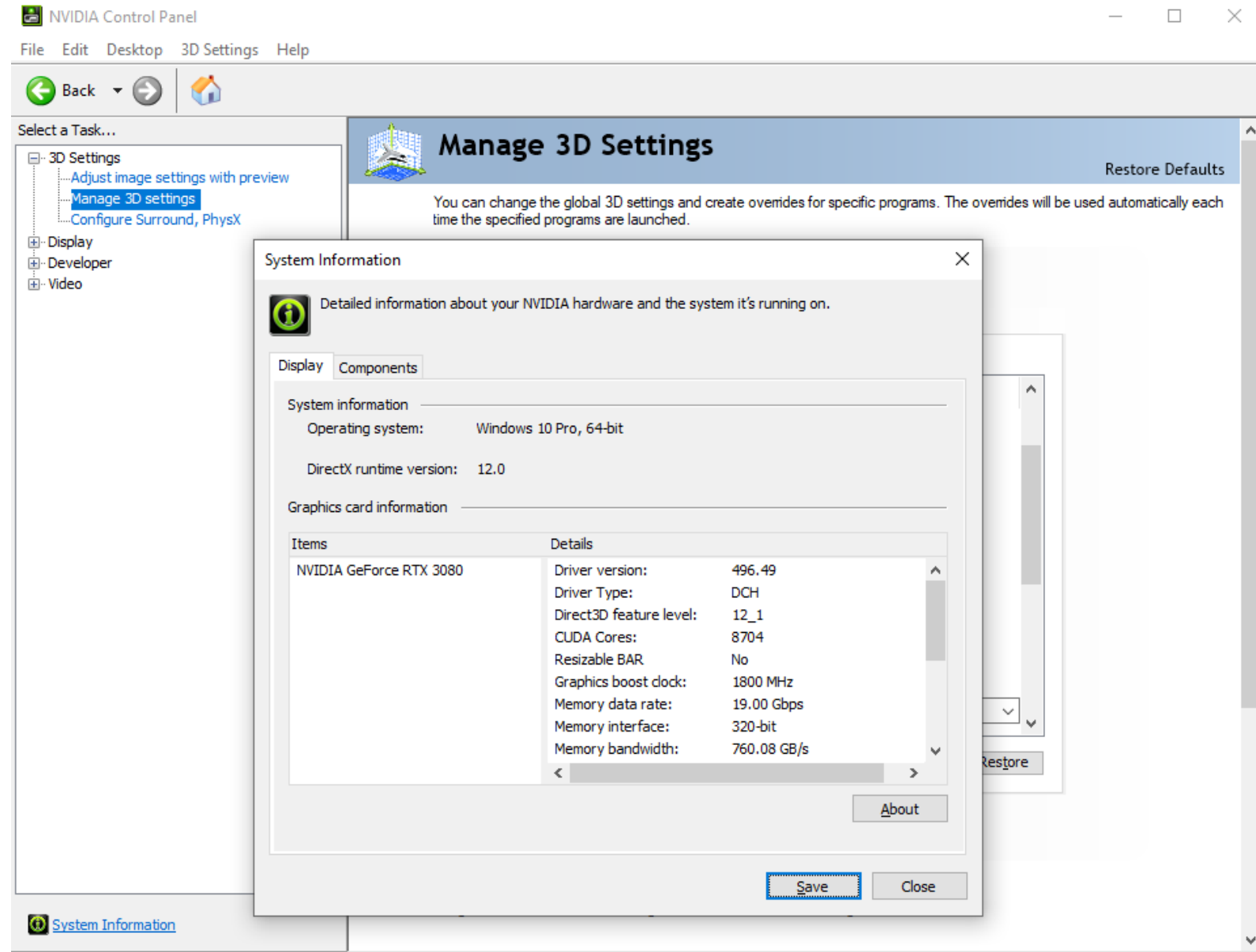
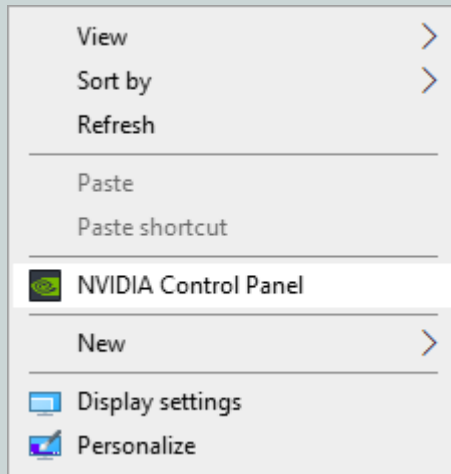




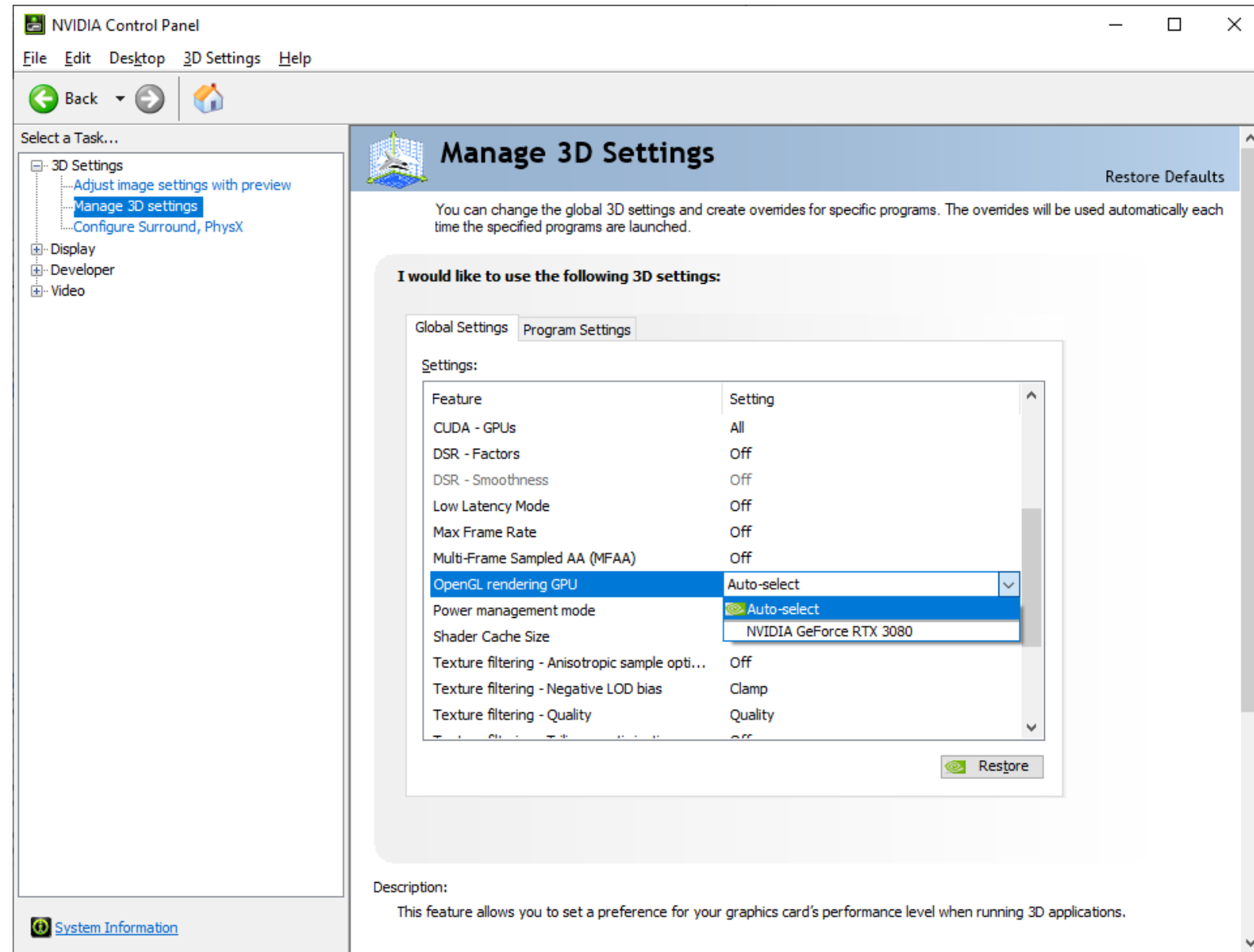
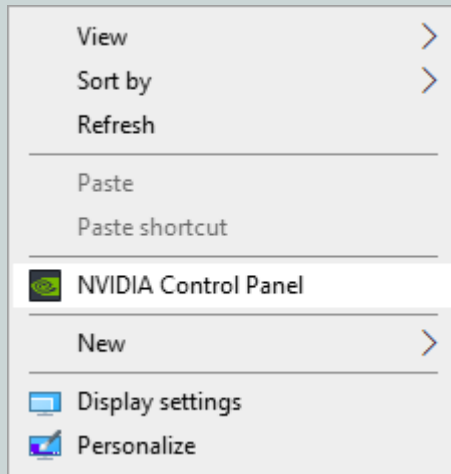
Go to your graphics card properties to check out which API is on your computer...



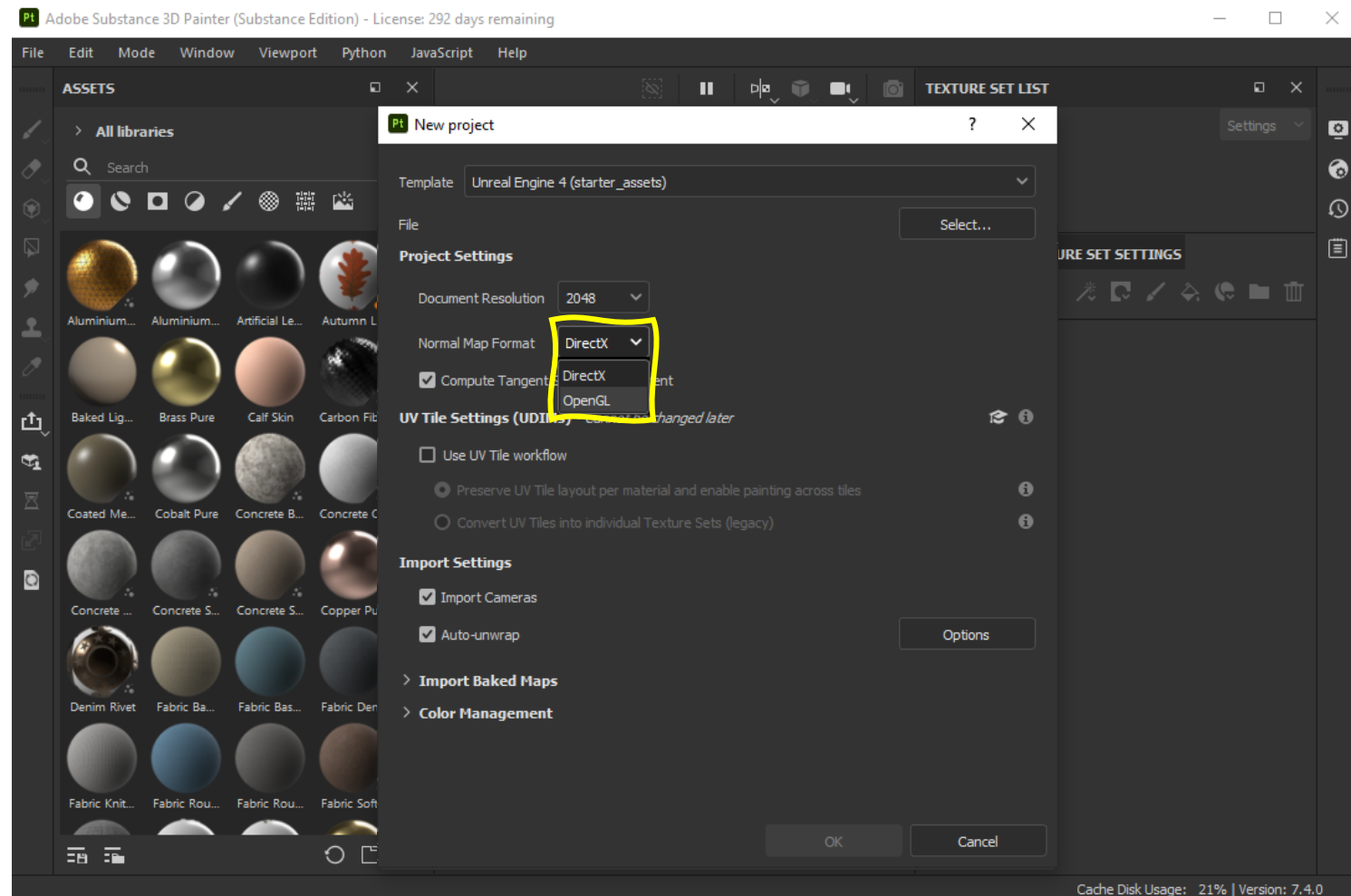
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Go to your graphics card properties to check out which API is on your computer...

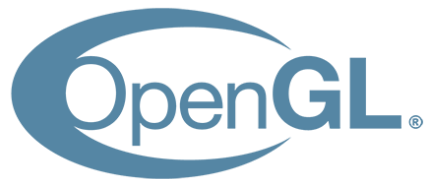


When you have more than graphics API on your computer, the operating system by default directs the apps to use one. Then Apps allow you to choose which one you want to go with.



GRAPHICS API

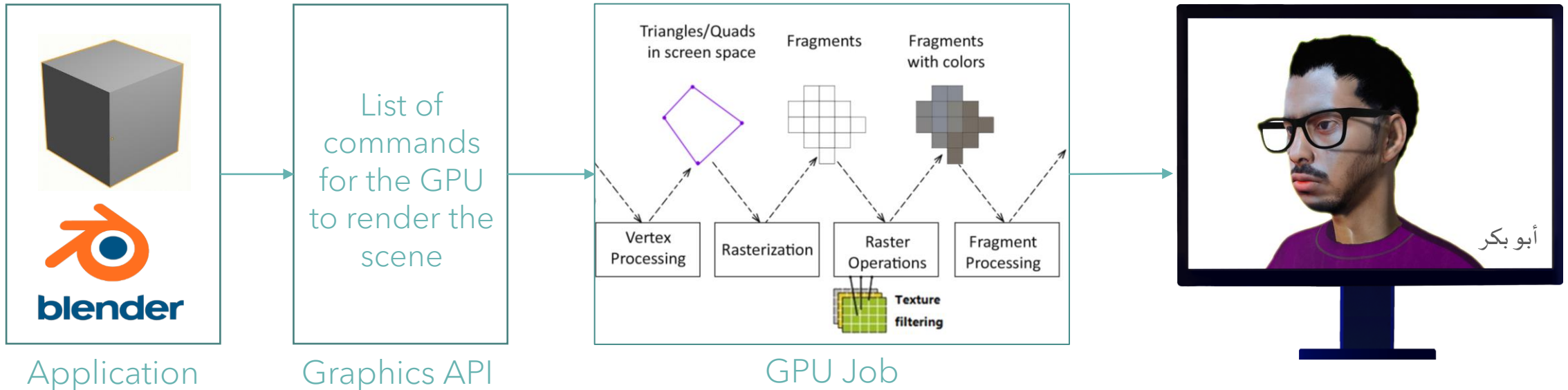
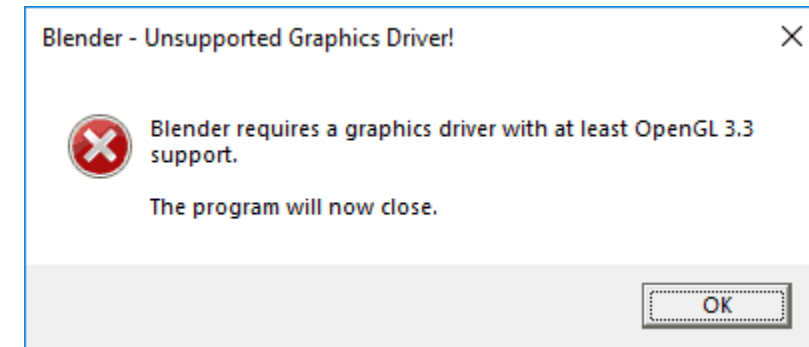
- If you are using Windows or Xbox, DirectX API is there. It is owned by Microsoft.
- OpenGL API is found on most computers regardless of the used operating system.
- OpenGL is a cross-language and cross-platform API. This is what makes it so popular.
- Vulkan API was introduced in 2016. It is developed by the non-profit Khronos Group, the same organization that maintains OpenGL.
- Other graphics APIs exist as well...



GRAPHICS API

How graphics API work?

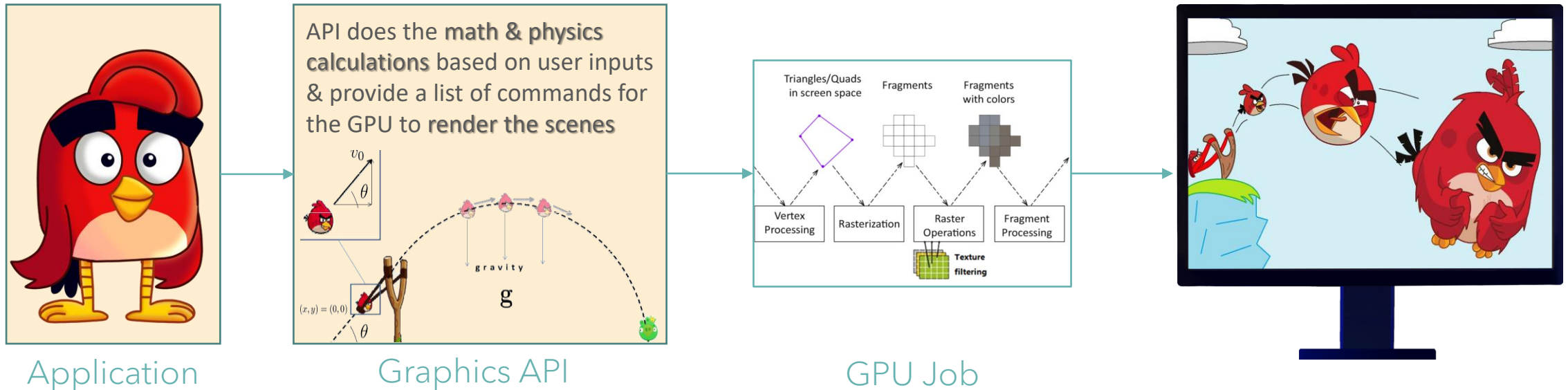
- Think of this example when you are using Blender to render a 3D scene
- Remember the restaurant example 😊



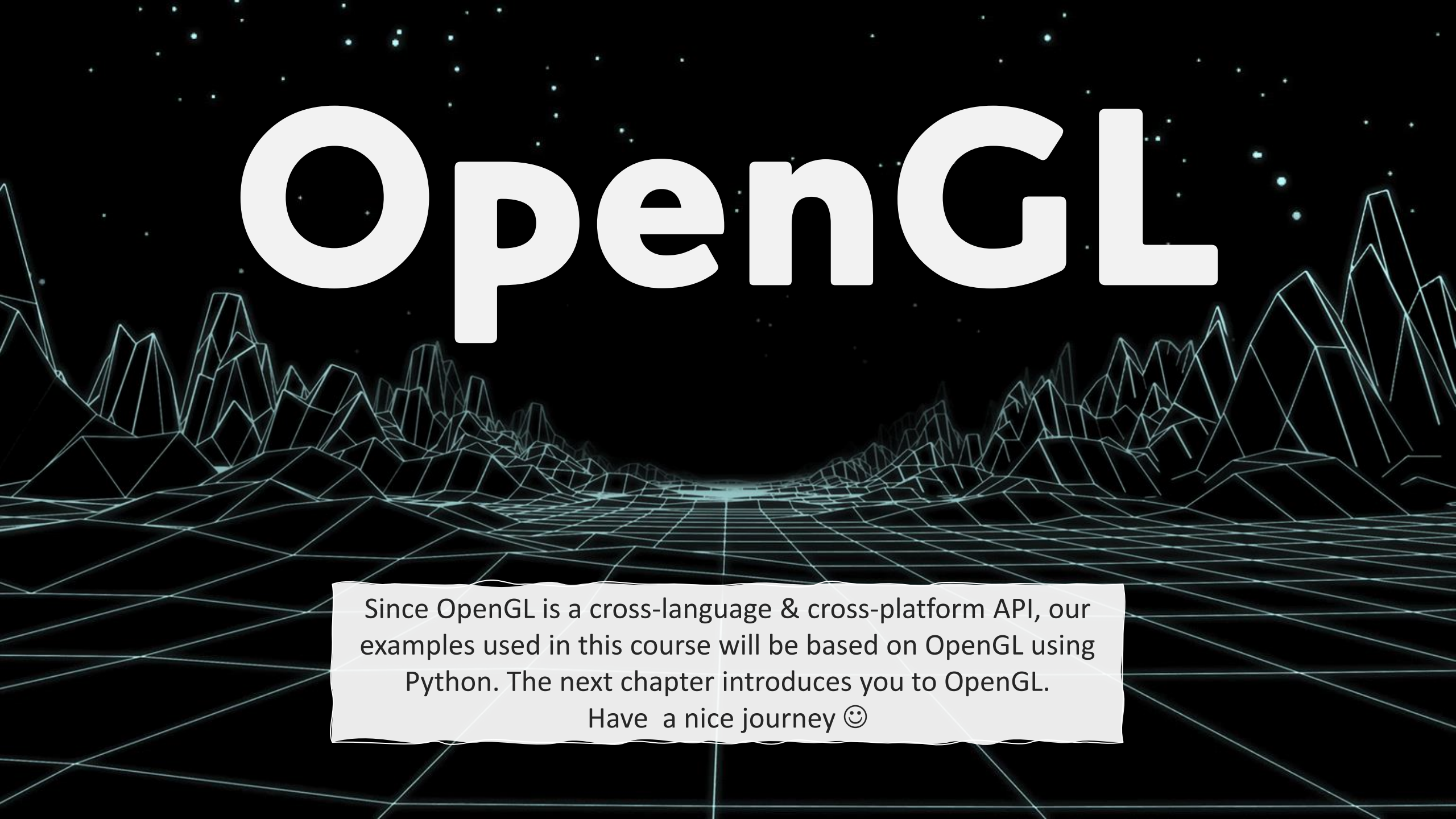
GRAPHICS API

How graphics API work?

- Think about a game
- A Game has animated objects that need to interact based on the user inputs...



OpenGL

The background of the slide is a wireframe landscape. It features a perspective grid on the ground that recedes into the distance. In the background, there are several jagged, mountain-like shapes also rendered in wireframe. The sky is black and filled with numerous small, white dots representing stars.

Since OpenGL is a cross-language & cross-platform API, our examples used in this course will be based on OpenGL using Python. The next chapter introduces you to OpenGL.

Have a nice journey 😊

QUESTIONS ABOUT GRAPHICS API

Can I make my own graphics API?

- Modern operating systems ban direct access to hardware for security reasons.

You don't want any random app to be able to read arbitrary GPU memory for the same reason you don't want any random app to be able to read system memory. Things like the frame buffer for your bank account or whatnot live in GPU memory. You want that stuff isolated and protected and access controlled by your OS.

Only drivers can talk directly to most hardware, and the only way to talk to the driver is via the OS' exposed HAL and each driver's proprietary and incompatible interface. That driver interface will not only be different for each vendor but will even differ between versions of the driver itself, making it near impossible to talk directly to the interface in a consumer application. These layers are often covered by access controls that further restrict the ability for an application to access them.

So no, your game cannot just use the hardware directly, unless of course you're only targeting insecure operating systems like DOS, and your only feasible option for a game on modern consumer operating systems is to target a public API like DirectX, OpenGL or Vulkan. ([Sean 2015](#))

QUESTIONS ABOUT GRAPHICS API

What is the difference between OpenGL & Blender/Maya?

- Is the difference the same as that between using machine language and a high-level language to write code? Is it lengthier to model using OpenGL than with Blender or Maya?

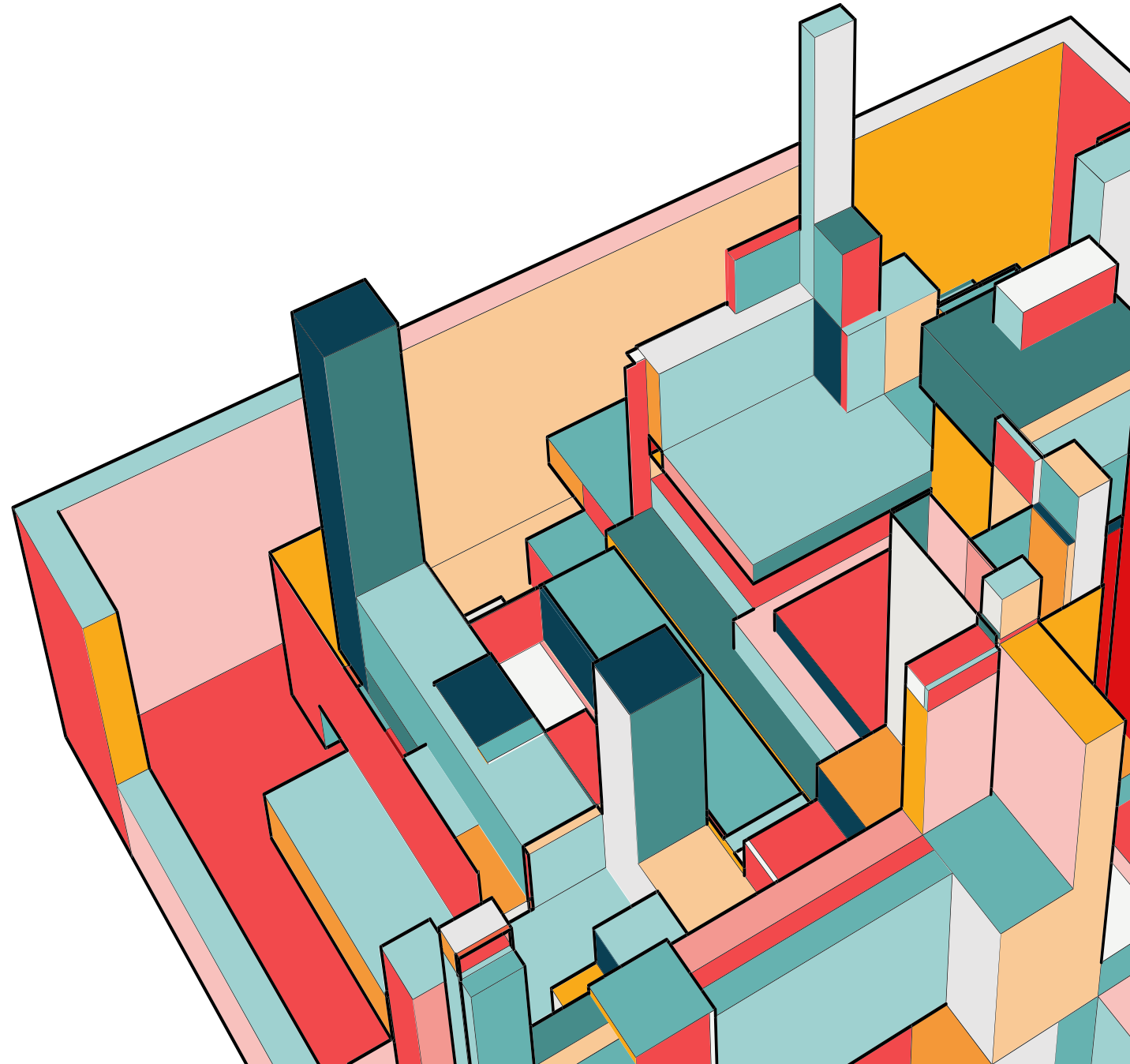
OpenGL provides an API that allows you to interact directly with your graphics card. Software like Blender and Maya add layers of abstraction on top of this, making them easier and more intuitive to use. The approaches are also different: whereas using OpenGL involves writing function calls in order to do things like render polygons to the screen, software like Maya and Blender are instead interacted with through a Graphical User Interface (GUI), where you create your objects using drags and drops...

OpenGL is used for taking data representing models created in another program (such as Maya or Blender) and rendering these to the screen. Maya and Blender still rely on OpenGL for actually rendering stuff on-screen (or DirectX could also be used), as do all programs which require drawing stuff on screen. (...[discussion on StackOverFlow](#))

100% CONFIDENT

Your journey in this course is
enjoyable and beneficial!

Enjoy learning CPIT285!



THANK YOU

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